Square-Wave Voltammetry Forward Model

How MMP-9 Concentration Becomes a Measured Voltammogram

Eton iGEM 2026, Dry Lab | Layers 8, 10, 11, 13, 15 | `swv_forward_model.py`

What this document is. This is the working model of how our electrode turns MMP-9 activity into a signal we can read on a potentiostat. It is organised as six self-contained parts, one per physical step in the chain from probe to peak. Each part follows the same pattern: what it does, in plain language; what we're assuming to get there, stated explicitly and justified; the equation, with its physical meaning given before it appears, not after; the parameters it needs; and the result. Where a number comes from a published paper it is marked ✓ confirmed. Where it is a working assumption, not yet confirmed on our own electrodes, it is marked ★ FLAG.

I. Overview: The Electrode and What Happens to It

Our sensor is a small gold electrode with a geometric area of 1.7 mm^2 (0.017 cm^2) [12] ✓ confirmed– confirmed against the fabricated device, superseding an earlier 1.5 mm-diameter design spec (0.0177 cm^2), a ∼4% difference too small to change any qualitative conclusion in this report. It carries two kinds of molecule stuck to its surface by a thiol (sulfur) anchor, Figure 1:

- **The signal probe.** A short peptide (SGKIPRTLTA) that MMP-9 cuts, with a spacer (PEG6) holding it away from the gold, and a methylene blue (MB) dye stuck on the far end. MB can pass an electron to the gold and back, which is what we detect. When MMP-9 cuts the peptide, the MB-carrying half floats away and that probe goes electrically silent.
- **The reference probe.** A short thiol carrying a ferrocene (Fc) group instead of MB. Fc is not cut by MMP-9 and gives its own, separate redox signal. We use its peak as an internal ruler to correct for how much probe actually ended up on a given electrode (Part E).

The two probes are deposited together in **one pot** (Part B). As MMP-9 activity increases, more MB is lost, and the MB peak in the voltammogram shrinks. Measuring how much it shrinks over a known incubation time is how we read out enzyme activity, and from there, estimate [MMP-9] in the wound sample.

The document is organised as six parts, each covering one physical step in that chain:

Part	Question	Section
A	What's actually on the electrode surface?	II
B	Where does the MB label sit above the gold?	III
C	How does one MB label's signal arise?	IV
D	Why square-wave voltammetry specifically?	V
E	How do we turn a peak into a cleavage fraction?	VI
F	How does [MMP-9] map to a predicted signal?	VII

Figure 0: Physical electrode assembly and tether geometry

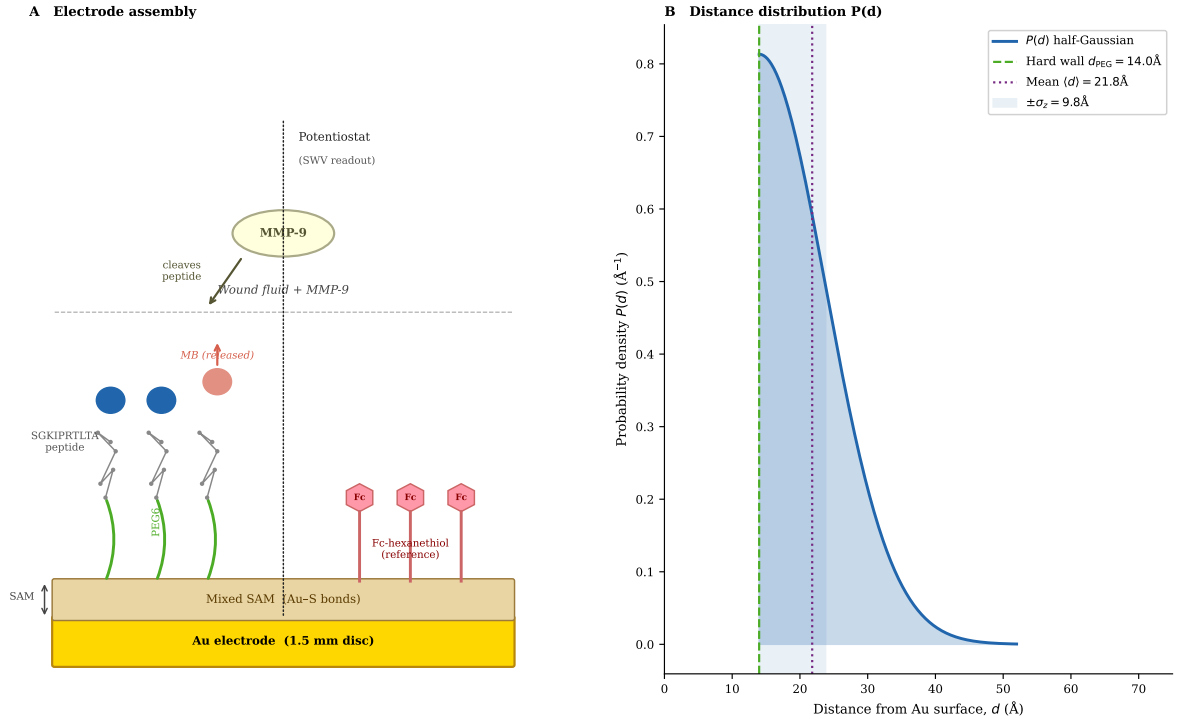


Figure 1: **Left:** the two probe types on the gold surface – the MB-peptide (cleavable) and the Fc reference (not cleavable). **Right:** the range of distances the MB label can sit at above the gold, set by the flexibility of its tether (Part III).

II. Part A: The Surface Chemistry

Overview

Before anything electrochemical can be modelled, we need to know what is physically on the electrode. **This section replaces an earlier, incorrect description.** Previous versions of this report and related layer reports described an MCH backfill step, where a small inert molecule (6-mercapto-1-hexanol) fills in gaps between the probes. **This is not what we do.** The current protocol [12] co-deposits both probes, MB-peptide and Fc-thiol, together in a single one-pot reaction. There is no separate MCH step, and no MCH at all. Fc plays the role MCH used to play in older versions of this model: it dilutes the MB-peptide probe density, spacing the signal probes apart.

Assumptions

- **One-pot, two-component deposition (no MCH).** Confirmed protocol [12]; this is a fact about the fabrication process, not a modelling choice.
- **Surface composition may not match the solution mixing ratio.** The peptide probe is thermodynamically favoured to stick to bare gold over the Fc-thiol, so it may end up over-represented on the surface relative to what was mixed in solution ★FLAG. This is an open question, not a design parameter we control precisely, and has not been measured.
- **Total probe coverage $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}.** This is *not* a measured value for our electrodes: it is an engineering placeholder in the range reported for a comparable Fc-peptide-PEG system in the literature [8] ★FLAG. This is not measurable with the instrumentation currently available to the team [12], so it remains a placeholder for the foreseeable future rather than a near-term fix; any prediction that depends sensitively on the exact value of Γ_{tot} should be treated with that in mind.
- **Bare gold exposure between probes.** Because there is no MCH filling the remaining surface, bare gold may be exposed between probes, which is a plausible route for non-specific protein adsorption from wound fluid (fouling). This was not a concern under the old MCH-backfill picture and has not been checked ★FLAG.

Why this matters downstream. The distance distribution $P(d)$ used in Part III was derived assuming a clean, PEG6-spaced probe with nothing else nearby. It has not been re-derived or validated for the real, MCH-free,

competitively-adsorbed surface described above. Treat $P(d)$ as a reasonable working shape, not a measured one.

III. Part B: Where Does the MB Label Sit?

Overview

The MB label is not rigidly fixed above the gold: it sits at the end of a flexible tether (PEG6 spacer plus the peptide itself), so it wobbles around in a range of distances. Molecules further from the gold transfer electrons much more slowly (electron transfer through the tether decays roughly exponentially with distance – this is standard Marcus/tunnelling theory [9,10]), so only probes sitting close to the gold contribute meaningfully to the signal.

Assumptions

Assumptions

- **The tether behaves as a freely-jointed chain (FJC).** This is a standard polymer-physics model for a flexible, non-rigid linker, used here for the peptide segment of the tether. It is a modelling choice, not a measurement on our specific peptide.
- **This geometry has not been validated on the real (MCH-free) surface.** See Part II: the calculation below assumes a clean, isolated PEG6-spaced probe, which may not hold once competitive peptide/Fc adsorption is accounted for ★FLAG.

Equation and parameters

Treating the tether as an FJC gives a distance distribution that peaks right at the PEG6 spacer length and falls off further out:

$$P(d) \propto \exp\left(-\frac{(d - d_{\mathrm{PEG}})^2}{2\sigma_z^2}\right), \qquad d \geq d_{\mathrm{PEG}}$$

with $d_{\mathrm{PEG}} \approx 14$ Å (from the PEG6 spacer) and $\sigma_z \approx 9.8$ Å (from the peptide's flexibility). The electron-transfer rate at the near end of that range works out to $k^0_{\mathrm{ref}} \approx 127\ \mathrm{s}^{-1}$ ★FLAG (carries the same $P(d)$ caveat above). At the team's confirmed operating frequency of 10–20 Hz [12], this gives $\kappa_{\mathrm{tip}} = k^0_{\mathrm{ref}}/(2f) = 6.4$ at 10 Hz or 3.2 at 20 Hz – both comfortably in the quasi-reversible range (Part IV) where signal is strongest, though the exact value depends on which frequency is actually used for a given scan. (An earlier version of this report used an assumed 15 Hz, giving $\kappa_{\mathrm{tip}} = 4.2$; this has been superseded by the confirmed frequency.)

Result

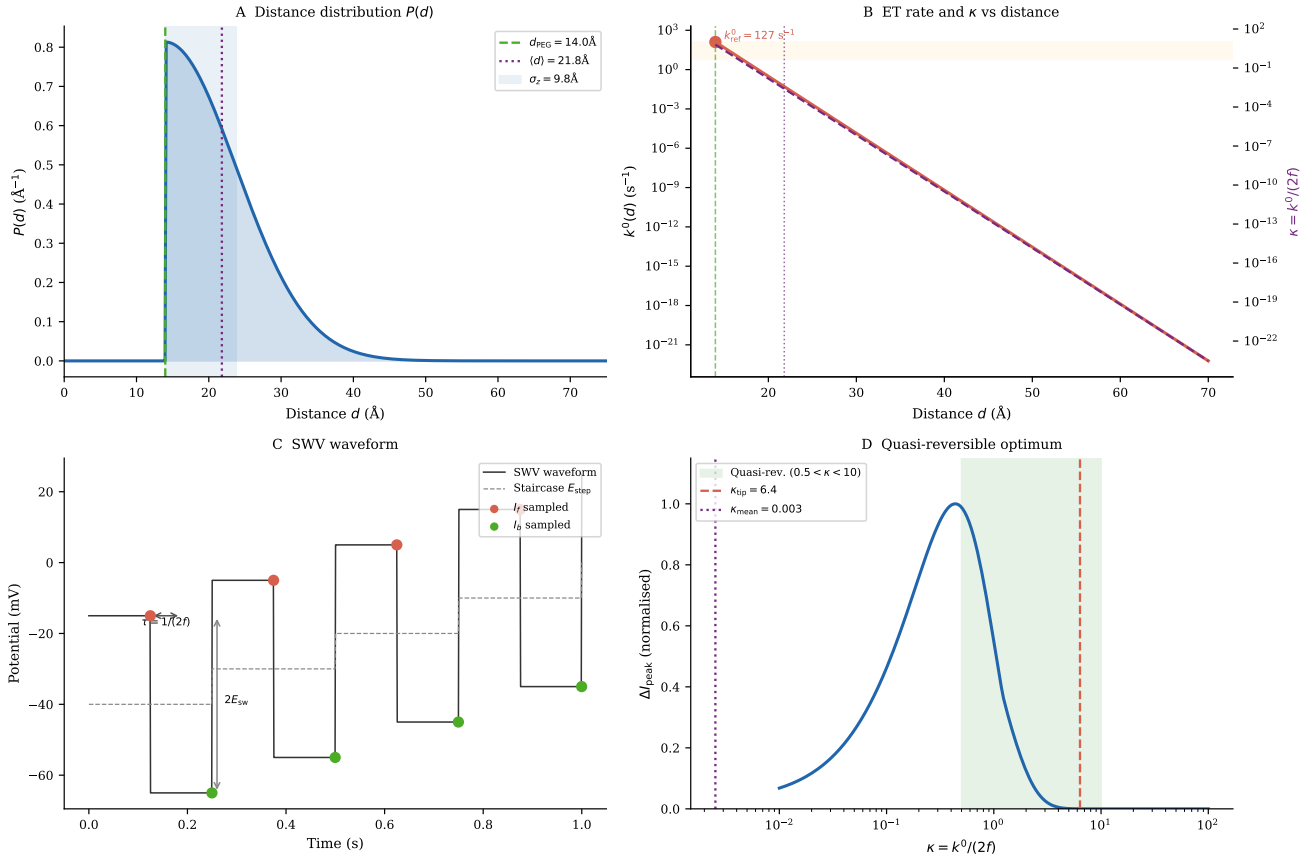


Figure 2: **A:** the distance distribution $P(d)$ – most MB labels sit within about 1 nm of the PEG6 spacer tip. **B:** the electron-transfer rate $k^0(d)$ falls off fast with distance, so nearby probes dominate the signal. **C:** the square-wave voltammetry (SWV) pulse pattern used to scan the electrode. **D:** the net signal peaks at an intermediate scan speed – too fast and probes can't react in time, too slow and forward/backward pulses give the same answer and cancel out.

IV. Part C: The Core Signal Model

Overview

Each MB label switches between an oxidised and reduced state at a rate set by the applied potential (Butler-Volmer kinetics, the standard model for electrode reactions [9]). Because MB is stuck to the surface rather than floating free in solution, this reduces to a single, exactly solvable equation – no numerical simulation needed – for the fraction of MB in the reduced state.

Assumptions

Assumptions

- **Butler-Volmer kinetics apply to surface-confined MB.** This is the standard, textbook model for a redox species attached to an electrode, not specific to our device, and is not in question here.
- **The potential is held fixed within each SWV half-pulse.** This is true by construction of the SWV waveform (Part V) and is what makes the equation below exactly solvable.

Equations

The rate of change of reduced-MB coverage Γ_{red} follows:

$$\frac{d\Gamma_{\text{red}}}{dt} = k_f\,(\Gamma_{\text{tot}} - \Gamma_{\text{red}}) - k_b\,\Gamma_{\text{red}} \tag{IV.1}$$

Because k_f, k_b are constant during a half-pulse, this has a closed-form solution:

$$\boxed{\Gamma_{\text{red}}(\tau) = \Gamma_{\text{eq}} + \left[\Gamma_{\text{red}}(0) - \Gamma_{\text{eq}}\right] e^{-(k_f + k_b)\tau}} \tag{IV.2}$$

In words: over each pulse, the MB coverage relaxes exponentially towards its equilibrium value Γ_{eq}. If the pulse is short compared to the reaction rate, almost nothing happens (weak signal). If it's long, the system fully equilibrates before the next pulse and forward/backward currents cancel (also weak signal). The useful signal comes from the middle ground, where the coverage moves partway to equilibrium each pulse – this is why there is an optimal scan frequency (Fig. 2D).

Scanning this across a full range of potentials, and adding up the contribution from every MB label at every distance d (weighted by $P(d)$ from Part III), gives the full predicted voltammogram $\Delta I(E)$ – a peak centred near the MB formal potential, whose height scales with how much intact probe remains on the electrode:

$$\Delta I(E; F_k) \;\propto\; (1 - F_k)\,\Gamma_{\mathrm{tot}} \tag{IV.3}$$

where F_k is the fraction of probe that has been cut by MMP-9. This proportionality was checked numerically and holds to within 2% across the full range of F_k – the peak shrinks in proportion to cleavage, and its position and width barely move.

Results

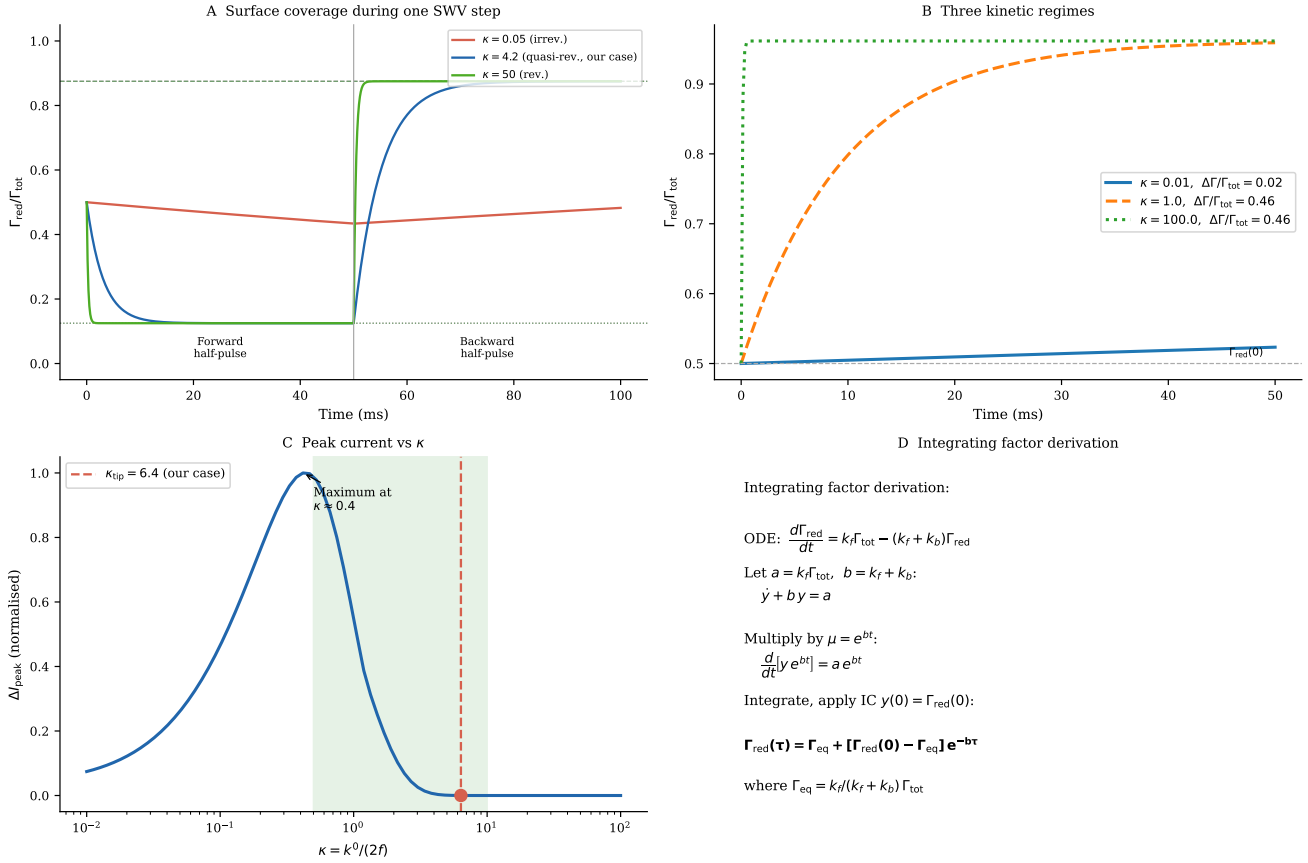


Figure 3: **A:** how the MB coverage relaxes during one forward-then- backward pulse pair, for three different reaction speeds. **B:** the three regimes – too slow, just right, too fast. **C:** net signal vs. reaction speed, confirming the optimum. **D:** the algebra behind Eq. (IV.2) (an integrating-factor solution of Eq. (IV.1)), for reference.

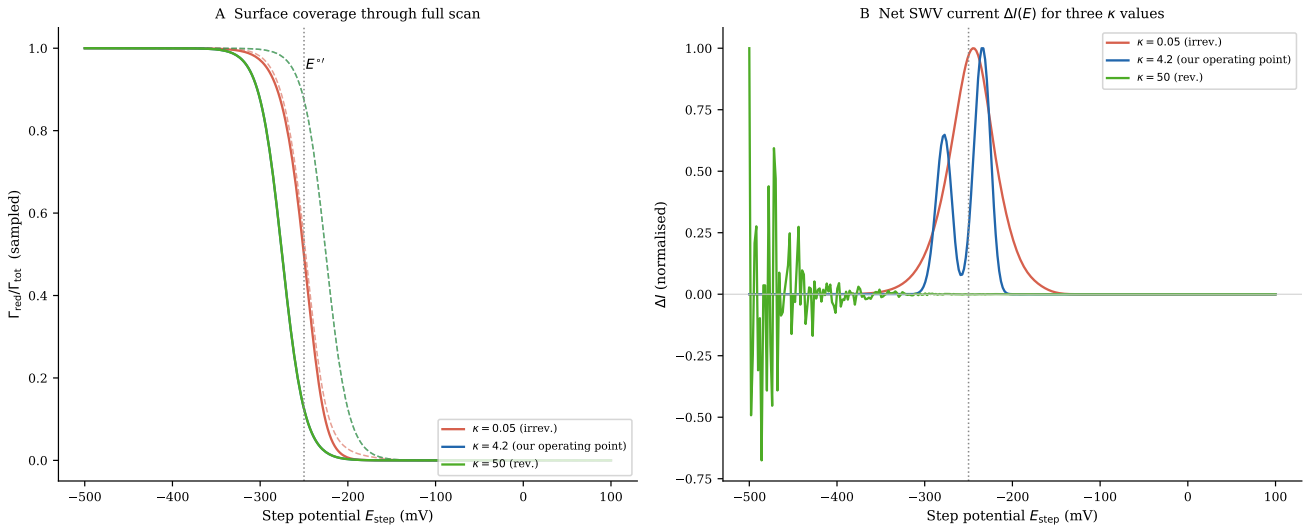


Figure 4: Predicted voltammograms scanning across the full potential range, for a fast, an intermediate, and a slow reaction speed (same three cases as Fig. 3). The intermediate case gives the sharpest peak.

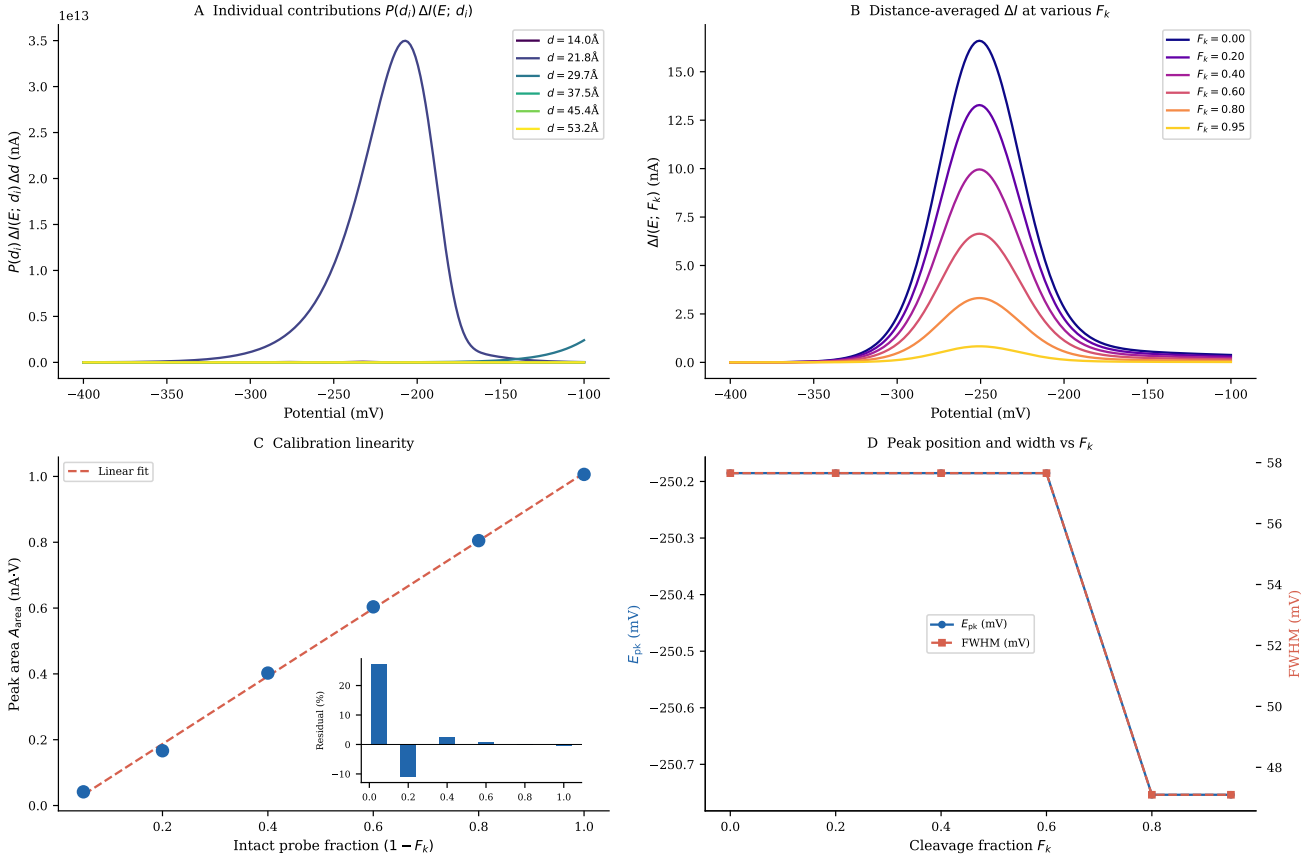


Figure 5: **A:** near probes dominate the total signal; far probes contribute almost nothing. **B:** predicted voltammograms at several cleavage fractions F_k – the peak shrinks but keeps its shape. **C:** peak area vs. $(1 - F_k)$, confirming the linear relationship of Eq. (IV.3). **D:** peak position and width stay essentially constant across F_k.

Validation against published data

Before trusting this model's predictions, we checked it against two independent published results. In the limit of a very slow scan, the model should reproduce the classical Laviron (1979) peak shape [2], which has a known, exactly-solvable width; our simulator matches it to under 1%. At faster scan speeds it should match measurements

from Dauphin-Ducharme et al. (2017) [1] on a comparable MB-on-gold system; our simulator's predicted peak width matches their measured values to within 4 mV across their tested frequency range. Both checks land in the same kinetic regime our device operates in, which gives reasonable confidence in the ODE solution itself, independent of the surface-chemistry open questions in Parts II–III.

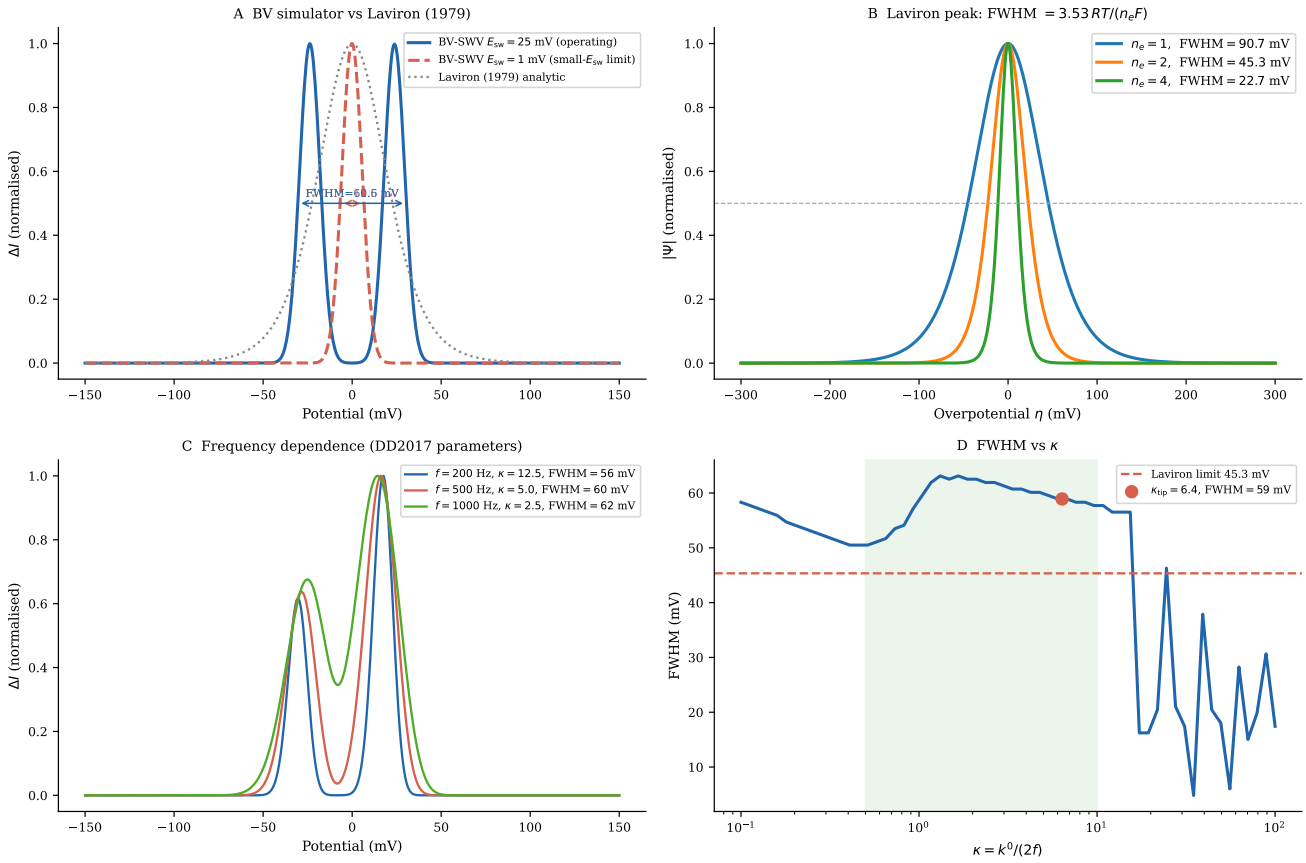


Figure 6: **A:** simulator vs. the exact Laviron (1979) peak shape at slow scan speed. **B:** the same comparison at three different numbers of transferred electrons. **C:** simulator vs. measured peak widths from Dauphin-Ducharme et al. (2017) across their frequency range. **D:** peak width vs. reaction speed, showing where our device sits relative to the validated range.

V. Part D: Why Square-Wave Voltammetry

Overview

Square-wave voltammetry samples the current only at the end of each pulse, by which point the electrode's background charging current has died away, and then subtracts the forward and backward readings. Both steps remove background signal that a simpler technique (e.g. cyclic voltammetry) would keep, giving SWV substantially better sensitivity for a surface-bound probe like ours [11].

Assumptions

Assumptions

- **SWV's sensitivity advantage over CV/DPV for surface-bound probes is standard electroanalytical theory** [11], not specific to our device.

Operating parameters

For completeness, the specific SWV settings this model uses: pulse amplitude $E_{sw} = 25$ mV [6,12] ✓ confirmed (matches the Baker 2006 literature value exactly, and confirmed against the fabricated device and potentiostat settings); operating frequency 10–20 Hz [12] ✓ confirmed; electrode area 1.7 mm^2 (Overview) ✓ confirmed; transfer coefficient $\alpha = 0.37$ and $n_e = 2$ for MB [1] ✓ confirmed; $\alpha = 0.5$ and $n_e = 1$ for Fc, the standard textbook assumption for a fast, symmetric one-electron couple rather than a measured value for this specific construct.

Result

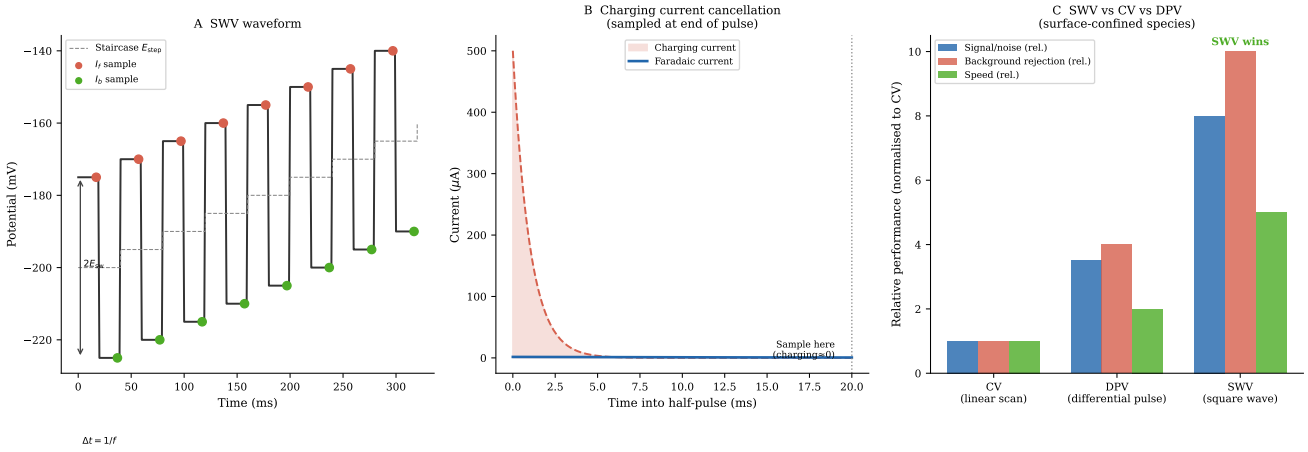


Figure 7: **A:** the SWV pulse sequence – a staircase with a square wave superimposed, forward and backward currents sampled at the end of each half-pulse. **B:** background charging current dies away quickly; the real (Faradaic) signal persists, so sampling late removes background. **C:** SWV's sensitivity advantage over CV and DPV for a surface-bound probe.

VI. Part E: The Ratiometric Correction

Overview: what it's trying to do, in plain terms

Every electrode we make ends up with a slightly different amount of probe on it. If we just measured the MB signal, a bigger MB peak might mean more MMP-9 cleaved less probe, or it might just mean that electrode happened to get more probe deposited in the first place. There's no way to tell those apart from the MB signal alone.

The fix is to also measure the Fc peak on the same electrode, in the same scan. Because Fc isn't cut by MMP-9, its peak size should track how much total probe landed on that electrode, regardless of enzyme activity. Taking the ratio of the two peaks is meant to cancel out the electrode-to-electrode variation and leave a number that depends only on how much MB was cleaved. That's the whole idea: compare the MB peak to the Fc peak before and after incubation, and any electrode-specific scaling factor cancels out of the ratio.

Equation

$$F_k^{\mathrm{rec}} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{Fc}}}{\Delta I_{\mathrm{MB}}(0)/\Delta I_{\mathrm{Fc}}} \tag{VI.1}$$

where ΔI_{MB} and ΔI_{Fc} are the net SWV peak *heights* (the ΔI_{peak} quantity used throughout Parts IV–V), not the integrated peak area used in earlier versions of this report – see the first assumption below for why that change matters.

Assumptions – and why this is currently an open problem, not a trusted result

This cancellation is an assumption, not a demonstrated fact for our device. Five separate problems have come up:

Assumptions

- **Peak height, not peak area, is the right quantity here – and even height isn't fully clean.** Earlier versions of this report defined the ratio using integrated peak *area* (height times width), by analogy with CV, where integrated charge is a physically meaningful quantity directly tied to coverage by Faraday's law. That analogy doesn't hold for SWV: the net SWV peak has no equivalent charge interpretation, and its width tracks the kinetic regime κ (Part IV), not coverage. On small, low-reproducibility SAM electrodes, fouling can change the electron-transfer kinetics at the Fc reference itself, not just the amount of Fc present – which distorts peak *shape* (width) independently of coverage. An area-based ratio cannot distinguish "coverage changed" from "kinetics/shape changed due to fouling," so it silently assumes fouling only ever scales peak amplitude. Peak height is closer to the standard SWV observable and more directly tied to coverage at the peak potential, but it is still affected by κ, so it is not immune to this problem either – distinguishing a real coverage change from a fouling-driven kinetics shift requires comparing height *and* width separately, not folding them into one number ★ FLAG. This has not yet been checked on our electrodes.
- **The Fc peak position doesn't match what the model expected.** We had modelled the Fc formal potential at +0.210 V vs. Ag/AgCl, taken from a comparable system in the literature [1]. Our own electrode measurements put it nearer **+0.300 to +0.400 V** [12] ★ FLAG – a 90–190 mV shift, and the value itself varies electrode-to-electrode rather than sitting at one fixed potential. This isn't fatal on its own, but it means the literature anchor doesn't transfer cleanly to our chemistry, and the real value needs a dedicated single-component Fc scan to pin down more precisely.
- **The borrowed Fc rate constant likely does not apply to this device either.** $k^0_{\mathrm{Fc}} = 2.7 \times 10^5$ s^{-1} was borrowed from a different construct (Dauphin-Ducharme 2017, MB-DNA electrode), the same way k^0_{ref} for MB was. At that rate and the team's confirmed operating frequency (10–20 Hz, not the 15 Hz this model previously assumed [12]), κ_{Fc} works out to 6,750–13,500 – so far into the reversible limit that the BV-ODE (Part IV) predicts an essentially zero net SWV signal, mathematically, regardless of coverage. That contradicts direct observation: a real Fc peak has been seen experimentally, with height varying from 5×10^{-8} to 10^{-5} A and width varying substantially between scans [12] ★ FLAG.

  Back-calculating from the worst-case observed height (5×10^{-8} A) at $f = 20$ Hz, using the model's own internal Fc coverage estimate $\Gamma_{\mathrm{Fc}} = 1.5 \times 10^{-12}$ mol cm^{-2} and the now-confirmed electrode area (1.7 mm^2) and SWV amplitude (25 mV), gives an optimal $k^0_{\mathrm{Fc}} \approx 23$ s^{-1} ($\kappa_{\mathrm{Fc}} \approx 0.6$, near the quasi-reversible optimum) – roughly four orders of magnitude slower than the borrowed value. At that corrected k^0_{Fc} and the current coverage assumption, the model cannot reach the upper end of the observed range (10^{-5} A) at *any* k^0: coverage, not kinetics, sets a hard ceiling on peak height, and 10^{-5} A is roughly 400× above that ceiling. One coherent (but unconfirmed) explanation is that real Fc coverage on a well-formed SAM is ∼400× higher than the current placeholder – which would also help explain the ∼100× electrode-to-electrode signal variation already noted above, if coverage varies similarly. **This is still not confirmed**: it depends on whether the observed heights were genuinely isolated Fc peaks rather than a combined or contaminated signal, which has not been checked. Electrode area and SWV amplitude are no longer open questions here (both confirmed above), so the remaining uncertainty is narrower than before, but this should still be treated as a specific, testable hypothesis, not a corrected parameter.
- **We have not actually seen two separate peaks.** The model assumes the MB and Fc peaks are well separated and can both be read off cleanly in one scan. **On real voltammograms, no distinct dual peak has been observed** ★ FLAG. We don't yet know if this is because the peaks genuinely overlap, because the MB signal is too weak to see next to the Fc signal at current coverages, or something else in the protocol. Until this is resolved (e.g. with probe-only and Fc-only control scans, or a peak-fitting approach to separate overlapping peaks), the ratiometric correction as written cannot actually be applied to a real scan.
- **The correlation the ratio depends on is unverified.** Equation (VI.1) only works if Fc coverage and MB coverage scale together electrode-to-electrode, since they're co-deposited. That's a reasonable assumption but has not been tested on our device, and preliminary data may not support it ★ FLAG. It should be tested directly: deliberately vary total probe loading and check whether the ratio still behaves as predicted.

Result

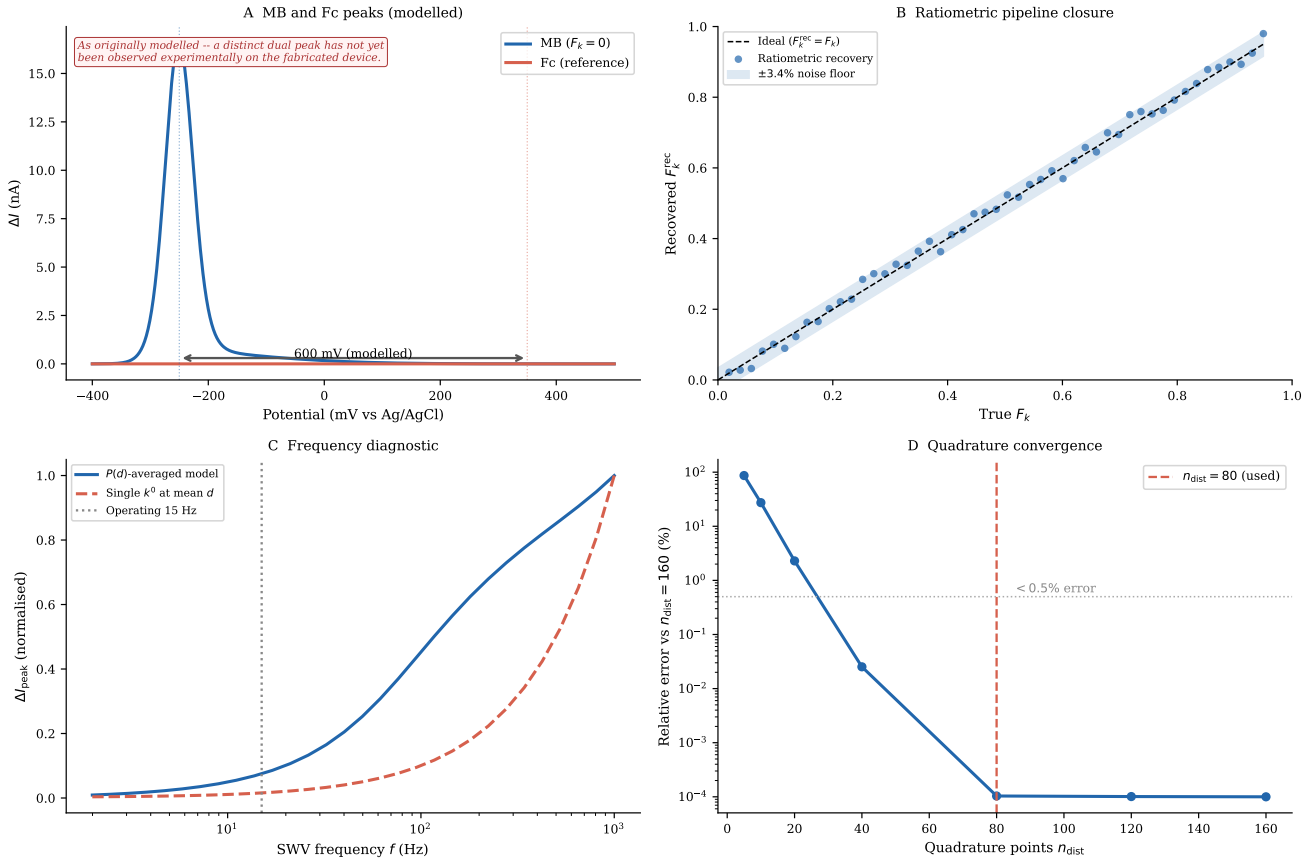


Figure 8: The ratiometric correction *as modelled*. **A:** MB and Fc peaks as modelled – the Fc trace is effectively flat at this scale. This is *not* a plotting error: using the literature k^0_{Fc} (see the assumption below), the model predicts a vanishingly small Fc signal at the team's operating frequencies, in contrast to the real Fc peak that has been observed experimentally – see the assumption below for what this implies. **B:** recovered vs. true cleavage fraction under an idealised noise model, independent of the area/height question. **C, D:** frequency and quadrature diagnostics for the underlying simulator (already peak-height-based), independent of the ratiometric open questions above.

A related concern: is a single calibration curve realistic?

Separately from the ratiometric question, the current plan is to convert a measured signal into [MMP-9] via a single calibration curve built from a reference set of electrodes. This works well for a large, reusable industrial electrode, where the same physical surface is measured repeatedly and stays consistent. Our electrodes are small and single-use, and electrode-to-electrode variation in probe coverage is already a known issue (Part II): measured signal amplitude has varied by up to a factor of 100 between nominally identical electrodes [12]. **A single fixed calibration curve is very unlikely to be robust to variation of this size** ★FLAG. This is the strongest argument in this report for relying on the ratiometric correction (Eq. VI.1) rather than a fixed calibration curve, once the dual-peak problem above is resolved – a within-electrode ratio is far less exposed to electrode-to-electrode variation than a curve built from a separate reference batch.

VII. Part F: From MMP-9 Concentration to Predicted Signal

Overview

At the enzyme concentrations expected in a wound sample (roughly 6–30 nM), MMP-9 is present in large excess over the surface probe, so the cleavage fraction follows simple first-order kinetics.

Assumptions

Assumptions

- **MMP-9 is in large excess over surface probe, so cleavage is pseudo-first-order.** Consistent with clinical concentration ranges (6–30 nM) against a surface probe density orders of magnitude lower.
- **Surface efficiency** $\eta = 0.20$**.** A central estimate, not a measured value for our device ★FLAG. It corrects for cleavage on a surface typically being slower than in free solution (steric hindrance, orientation effects), and still needs to be measured directly (e.g. by comparing electrode readout to a parallel solution-phase fluorescence assay).

Equation and parameters

$$F_k(T) = 1 - e^{-k_{\text{eff}}\,[\text{MMP-9}]\,T} \tag{VII.1}$$

where $k_{\text{eff}} = \eta \times (k_{\text{cat}}/K_m)_{\text{sol}}$, and $(k_{\text{cat}}/K_m)_{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ is the measured solution-phase cleavage rate for this exact peptide [7] ✓ confirmed.

Results

Wound state	[MMP-9]	F_k at 30 min ($\eta = 0.20$)	Source
Healing	6 nM	14%	Trengove 1999 [7b]
Chronic	30 nM	53%	Trengove 1999 [7b]

At the 3σ detection criterion, this gives a predicted limit of detection around **4 nM** at 30 minutes, below the 6 nM healing/chronic boundary – *provided* the ratiometric noise floor used in that calculation (∼3%) is achievable, which depends on resolving the dual-peak issue in Part VI first.

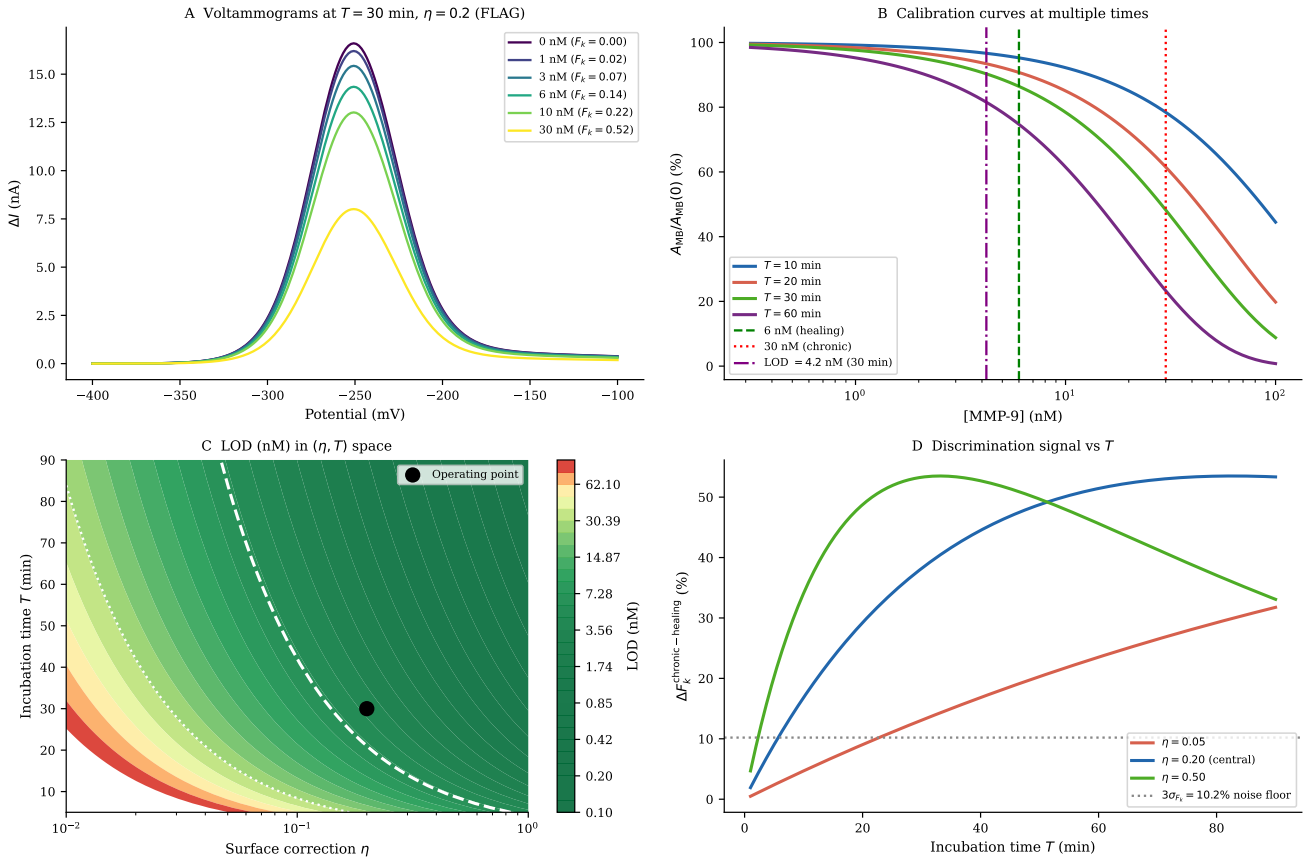


Figure 9: **A:** predicted voltammograms across a range of clinically relevant [MMP-9]. **B:** calibration curves at several incubation times. **C:** predicted detection limit as a function of the surface efficiency η and incubation time T. **D:** predicted signal difference between healing and chronic wound states vs. incubation time.

VIII. Summary: What's Solid, What's Assumed, What's Open

Part	Component	Status	
C, D	The BV-ODE and its solution	Exact maths, validated against two published datasets	✓ confirmed
D	SWV beats CV/DPV for this probe	Standard, well-established result	✓ confirmed
F	Solution cleavage rate (Kridel)	Directly measured for this peptide	✓ confirmed
A	One-pot, no-MCH deposition	Confirmed current protocol [12]	✓ confirmed
A	Operating frequency (10–20 Hz)	Confirmed by team; supersedes earlier 15 Hz assumption	✓ confirmed
B	$P(d)$ tether geometry	Derived for a clean surface, not validated for the real one	★ FLAG
A	Surface coverage Γ_{tot}	Engineering placeholder; not measurable with available instrumentation	★ FLAG
A	Electrode area (1.7 mm^2)	Confirmed against fabricated device (supersedes 1.5 mm design spec)	✓ confirmed
D	SWV amplitude E_{sw} (25 mV)	Confirmed against potentiostat settings; matches literature exactly	✓ confirmed
E	C6FC formal potential	Literature value doesn't match observed +300–400 mV, itself variable	★ FLAG
E	Fc rate constant k^0_{Fc}	Back-calculated ∼23 s^{-1}, ∼4 orders below borrowed value; unconfirmed	★ FLAG
E	Two resolvable peaks (MB, Fc)	**Not observed on real scans yet**	★ FLAG
E	Ratiometric drift-cancellation	Assumed, not demonstrated on our device	★ FLAG
E	Single calibration curve	Unlikely to be robust: up to 100× electrode-to-electrode variation	★ FLAG
F	Surface efficiency η	Central estimate, needs direct measurement	★ FLAG

Bottom line. The core electrochemistry (Parts IV and V) rests on standard, validated theory and isn't in question. What needs direct experimental attention before the device can be trusted end to end is everything to do with the real surface: what's actually on it (Part II), whether the two reporter peaks can even be told apart (Part VI), and whether a single calibration curve is the right strategy for electrodes this small and this variable (Part VI).

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